

INSERTABLE INTERNAL GEOMETRY/COMPLIANCE DEVICE — HARDWARE ARCHITECTURE, ACTUATION OPTIONS, POWER/COMMS, AND STANDALONE USE CASES (NON-LIMITING)

A. Hardware Architecture and Sensing Regions (Non-Limiting)

In various embodiments, Node 4 comprises a sealed insertable body containing electronics, a power source, and one or more sensing regions arranged to characterize internal distance, contact distribution, and mechanical response during controlled stimulus. In certain embodiments, the device includes: 1. one or more optical endpoint sensing regions (e.g., IR ToF) positioned to estimate internal distance; 2. a stimulus region comprising an expandable element (e.g., annular cuff/microbladder ring) configured to apply controlled radial stimulus; 3. a stimulus pressure sensing region comprising a MEMS pressure transducer configured to measure inflation/expansion pressure; 4. one or more circumferential contact sensing regions comprising piezoresistive/FSR rings, capacitive pressure rings, and/or piezoresistive film bands; 5. capacitive proximity/contact confirmation electrodes configured to confirm coupling; 6. a 6-axis IMU configured to capture motion/orientation for robustness and artifact rejection; and/or 7. one or more temperature sensors configured to provide thermal context and drift compensation. Sensing elements may be distributed and described as sensing regions rather than fixed placements to support design variation.

B. Actuation and Stimulus Mechanisms (Non-Limiting)

In various embodiments, controlled stimulus is produced by one or more actuation options including, without limitation: 1. manual pneumatic actuation via a compressible handle element configured to inflate an annular cuff/microbladder ring; 2. a micro-pump configured to provide controlled inflation profiles; 3. a mechanical cam or screw mechanism configured to generate controlled radial expansion; 4. a constraint band configured to shape expansion response; and/or 5. a hybrid mechanism combining manual actuation with regulated control via pressure sensing. In certain embodiments, the device includes mechanical stops or software limits to constrain maximum expansion and maximum pressure.

C. Sealing, Materials, and Cleaning (Non-Limiting)

In various embodiments, the device comprises a compliant outer shell (e.g., silicone or other biocompatible elastomer) molded over internal components with a sealed architecture to resist fluid ingress. In certain embodiments, the device is configured for cleaning using non-limiting procedures consistent with the housing and outer shell materials.

D. Power, Charging, and Communications (Non-Limiting)

In various embodiments, the device includes a rechargeable battery and power management circuitry configured for session-based operation. Non-limiting approaches include: 1. docking-based charging; 2. contact-based or inductive charging; 3. secure wireless communications (e.g., BLE) to transfer session data; and 4. firmware update mechanisms with integrity checks. In certain embodiments, data is buffered on-device and synchronized after the session.

E. User Guidance and Safety Constraints (Non-Limiting)

In various embodiments, a computing system provides guidance prompts and enforces limits. Non-limiting features include: 1. prompts to reduce motion when IMU gating indicates excessive movement; 2. prompts

to re-seat when coupling integrity is low; 3. prompts to pause or abort stimulus if pressure thresholds are exceeded; and 4. prompts to repeat a session when quality indices indicate reduced confidence.

F. Standalone Use Cases (Non-Limiting)

In standalone embodiments, Node 4 operates as an independent product to generate privacy-preserving internal geometry, compliance, and contact response references for a single user. Non-limiting standalone outputs include: 1. internal distance indices and distance range descriptors; 2. compliance indices and response curve descriptors; 3. zone-based contact distribution signatures; 4. repeatability indices across multiple sessions; and 5. confidence indicators based on coupling integrity, motion stability, stimulus plausibility, and sensor coherence.

G. Optional Integration Hooks and Variations (Non-Limiting)

In optional multi-node embodiments, Node 4 outputs may be time-aligned and correlated with other nodes under consent gating and quality thresholds, including correlation with external baseline geometry references and longitudinal physiology references. Any embodiment described herein may omit sensors, add sensors, substitute equivalent modalities, alter housing geometry, alter actuation mechanisms, and/or alter processing distribution (on-node, on-device, cloud) while maintaining the functional intent of controlled stimulus-based internal geometry and compliance characterization.